

$$6.5 + 1 =$$

7.5

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Álgebra Lineal

Grupo 17

Semestre 2022-2

### Examen parcial no. 2

1) a)

$$F(G(x, y)) = F(x, -y)$$

$$F(x, y) = (2x, 3y)$$

$$F(G(x, y)) = (2(x), 3(-y))$$

$$G(x, y) = (x, -y)$$

$$F \circ G = (2x, -3y)$$

b)

$$M(T) = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

$$|M(T) - \lambda I|$$

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & -3-\lambda \end{vmatrix}$$

$$(2-\lambda)(-3-\lambda) + (0) + (2-\lambda)(-3-\lambda) + (0) + (-3-\lambda)(2-\lambda) + (0)$$

$$-6 + \lambda + \lambda^2 - 6 + \lambda + \lambda^2 - 6 + \lambda + \lambda^2$$

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot (-6)}}{2 \cdot 1}$$

$$\lambda_{1,2} = \frac{-1 \pm 5}{2 \cdot 1}$$

$$\lambda_1 = \frac{-1 + 5}{2} = 2$$

$$\lambda_2 = \frac{-1 - 5}{2} = -3$$

$$\lambda_1 = 2$$

$$\lambda_2 = -3$$

→ Valores propios

Para  $\lambda_1 = 2$

$$(M(T) - \lambda I)[\vec{v}]_B$$

$$\begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -5 \end{pmatrix}$$

$$[\vec{v}]_B$$

$$[(x, y)]_B$$

$$(x, y) = \alpha_1(1, 0) + \alpha_2(0, 1)$$

$$(x, y) = (\alpha_1, \alpha_2)$$

$$[\vec{v}]_B = [(x, y)]_B = \begin{pmatrix} x \\ y \end{pmatrix}_H$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} -5y &= 0 \\ y &= 0_H \end{aligned}$$

$$x = x_H$$

$$V.C. = (\lambda_1 = 2) = \{(x, 0) \mid x \in \mathbb{R}\}$$

Vector característico de  $\lambda_1 = 2$

Para  $\lambda_2 = -3$

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} - (-3) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$5x = 0$$

$$x = 0$$

$$y = y$$

$$V.C. = (\lambda_2 = -3) = \{ (0, y) \mid y \in \mathbb{R} \}$$

Valor característico cuando  $\lambda_2 = 3$

c)

$$F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$G: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

Ambas tienen 2 bases

$$UB = \{ (1, 0), (0, 1) \}$$

$$M(T) = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

$$\dim B = 2 \text{ bases}$$

Matriz  $2 \times 2$

$\therefore$  Si existe una matriz diagonal que represente a  $F$  o  $G$

Nombre:

Día

Mes

Año

Folio

Tema:

d)

$$UB = \{ (1, 0), (0, 1) \}$$

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\left( \begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array} \right)$$

$$\therefore P^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$D = P^{-1} M(T) P$$

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

2)

$$T(a, b, c) = a + (a+b)x + (b+c)x^2$$

$$A = \{x^2, x, 1\}$$

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T: \mathbb{R}^3 \rightarrow P_2$$

$$M_B^A(T) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{matrix} \rightarrow x^2 \\ \rightarrow x \\ \end{matrix}$$

esta  
no  
es

$$\det M_B^A(T) = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 1$$

$$1 \neq 0 \quad \therefore \text{Si } \exists T^{-1}$$

$$M_B^A(T) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 0 & 1 & 1 & | & 1 & 0 & 0 \\ 1 & 1 & 0 & | & 0 & 1 & 0 \\ 1 & 0 & 0 & | & 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_3 - R_2 \\ R_2 \end{matrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & | & 0 & 1 & 0 \\ 0 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & -1 & 0 & | & 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & -1 & 0 & | & 0 & -1 & 1 \end{pmatrix}$$

Nombre: 

Día

Mes

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Tema: 

$$(-1)R_2$$

$$\left( \begin{array}{ccc|ccc} 1 & 0 & -1 & -1 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 1 \end{array} \right)$$

$$R_3 - R_2 = R_3 \quad (-1)R_2$$

$$\left( \begin{array}{ccc|ccc} 1 & 0 & -1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right)$$

$$(-1)R_3 \quad R_1 - R_3 = R_1$$

$$\left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{array} \right)$$

$$(-1)R_3 \quad R_2 - R_3 = R_2$$

$$\left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right)$$

$$M_A^B(T^{-1}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

## Teorema

$$M_A^B(T^{-1})[\bar{w}]_B = [T^{-1}(\bar{w})]_A$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{P}_2$$

$$T^{-1}: \mathbb{P}_2 \rightarrow \mathbb{P}^3$$

$$\bar{w} = (a, b, c)$$

$$B = \left\{ \underbrace{(1, 0, 0)}_{w_1}, \underbrace{(0, 1, 0)}_{w_2}, \underbrace{(0, 0, 1)}_{w_3} \right\}$$

$$\bar{w} = \alpha_1 \bar{w}_1 + \alpha_2 \bar{w}_2 + \alpha_3 \bar{w}_3$$

$$(a, b, c) = \alpha_1 (1, 0, 0) + \alpha_2 (0, 1, 0) + \alpha_3 (0, 0, 1)$$

$$(a, b, c) = (\alpha_1, \alpha_2, \alpha_3)$$

$$[\bar{w}]_B = (a, b, c) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} c \\ b - c \\ a - b + c \end{pmatrix}$$

$$\therefore T^{-1}(ax^2 + bx + c) = (c, b - c, a - b + c)$$